Understanding Birth Weight Variation in U.S. Birth Records: Individual Predictors and Subpopulation Heterogeneity

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ABSTRACT
This memo studies variation in birth weight using U.S. birth records, with emphasis on both individual predictors and broader subpopulation heterogeneity. I first examine how infant sex, maternal age, paternal age, plurality, and birth order relate to birth weight using regression models that allow nonlinear effects and county-level clustering. I then quantify heterogeneity across subpopulations defined by county, year, sex, and maternal race. The results indicate that birth weight is associated with several parental and infant characteristics, that some effects are nonlinear and non-additive, and that within-county dependence is present but modest after adjustment. Across subpopulations, birth weight shows measurable heterogeneity, while parental ages vary even more strongly. These findings suggest that birth weight variation is shaped by both individual-level demographic factors and broader population structure.

1 INTRODUCTION

Birth weight is a widely used indicator of infant health because it reflects conditions present before and at delivery and is closely tied to early-life risk. At the same time, birth weight is not determined by a single factor. It is shaped by a combination of infant characteristics, parental characteristics, and broader demographic and geographic context. This makes it a useful setting for statistical analysis: the outcome is continuous, large-scale birth record data are available, and both individual-level and group-level variation are likely to matter.

A natural first question is how birth weight changes with observed predictors such as infant sex, maternal age, paternal age, plurality, and birth order. These variables are scientifically relevant, but their effects may not be simple. For example, parental age effects may be nonlinear, and the association of one predictor with birth weight may depend on another predictor rather than remaining constant across the population. In addition, births recorded within the same county may share common social, environmental, or healthcare-related conditions, so an analysis that treats all observations as fully independent may understate uncertainty.

Beyond individual-level associations, it is also important to ask how much variation exists across broader subpopulations. Even after accounting for observed predictors, birth outcomes may differ systematically across counties, years, sexes, and racial groups. Studying this heterogeneity helps distinguish variation that arises at the subject level from variation that reflects broader population structure. It also provides a useful connection between regression modeling and data-integration ideas, since subgroup summaries can reveal patterns that may not be obvious from individual-level models alone.

2 DATA AND VARIABLES

The analysis uses U.S. natality records provided in the course materials. The individual-level regression analysis is based primarily on the 1971 birth records, where each observation corresponds to a single birth. The main response variable is birth weight, measured in grams. Key covariates include infant sex, maternal age, paternal age, plurality, and birth order. Geographic location is identified using state and county, which are combined to define county-level groups for clustered analyses.

Several variables contain nontrivial missingness, especially those involving paternal information. In the exploratory stage, missingness patterns showed that some variables, such as paternal age and paternal race, tend to be missing together, suggesting that the missing-data structure is not completely random. For the main regression analysis, I therefore worked with a complete-case

subset based on the variables included in the fitted models. I also recoded a few variables for stability, including capping very large values of birth order and centering maternal and paternal age before introducing nonlinear terms. These steps were intended to improve interpretability and reduce the influence of sparse or extreme values.

To study broader heterogeneity, I also use a multi-year version of the birth data that combines records from 1971, 1981, 1991, and 1992. In this part of the analysis, the focus shifts from individual observations to subpopulations defined by county, year, sex, and maternal race. For each subgroup, I examine summary measures such as the mean and standard deviation of birth weight, maternal age, and paternal age. In some parts of the subgroup analysis, I restrict attention to singleton first births so that strong mechanical effects of plurality and birth order do not dominate the comparisons across groups.

The variables included in the analysis were chosen to reflect both biological and demographic sources of variation. Birth weight is the primary outcome of interest, while infant sex and parental ages capture subject-level demographic structure. Plurality and birth order are included because they are known to be strongly associated with birth outcomes and may also modify the interpretation of other predictors. County, year, sex, and maternal race define a natural set of higher-level groupings for studying whether variation in birth weight persists beyond the individual level.

3 METHODS

3.1 Individual-level regression analysis

To study how birth weight is related to infant and parental characteristics, I fit regression models with birth weight as the response and infant sex, maternal age, paternal age, plurality, and birth order as predictors. Because birth weight is positive and somewhat right-skewed, I modeled the conditional mean using generalized estimating equations with a Gamma mean-variance relationship and a log link. County was used as the clustering unit, so that standard errors would account for dependence among births occurring in the same county rather than treating all births as independent.

Maternal and paternal age were allowed to enter the model nonlinearly through quadratic terms after centering. This allows the fitted mean birth weight to vary with age in a curved rather than purely linear way, which is important because both very young and older parents may be associated with lower birth weight. I began with relatively simple additive specifications and then expanded the model to allow selected interaction terms, with the goal of assessing whether the role of one predictor differs across levels of another predictor.

3.2 Model comparison and interpretation of non-additivity

To determine whether more flexible specifications were supported by the data, I compared nested generalized estimating equation models using score tests. This strategy makes it possible to assess whether adding nonlinear terms or interaction structure improves fit without relying only on coefficient-by-coefficient summaries. The progression of models moved from simpler additive forms toward richer specifications that allowed the relationship between birth weight and predictors such as maternal age, plurality, and birth order to vary jointly.

The main purpose of this stage was not only to identify statistically supported predictors, but also to determine whether an additive description is adequate. If a more flexible model fits better, this suggests that the fitted slope for one predictor is not constant across the values of another predictor. In that setting, interpretation focuses on how the predicted mean birth weight changes over combinations of predictors, rather than on isolated coefficients alone.

3.3 Subpopulation-level heterogeneity

To study broader population structure, I aggregated the multi-year data into subpopulations defined by county, year, infant sex, and maternal race. Within each subpopulation, I calculated the sample mean, sample standard deviation, and sample size for birth weight, maternal age, and paternal age. I restricted this part of the analysis to singleton first births so that the strong and mechanically understood roles of plurality and birth order would not dominate the comparisons across groups.

These subgroup summaries were used to assess how much variation occurs between subpopulations rather than within them. For each outcome, I estimated the intraclass correlation coefficient using a moment-based decomposition of variance. In this framework, the variance of the subgroup means is separated into a structural between-group component and a within-group estimation component, where the latter is captured through the squared standard errors of the subgroup means. This produces a debiased estimate of between-group variation and avoids overstating heterogeneity simply because some subgroup means are estimated with noise.

3.4 Quantifying between-group variation

The intraclass correlation coefficient was used as a scale-free summary of heterogeneity, interpreted as the proportion of total variance attributable to differences between subpopulations. Because this proportion can be difficult to interpret on its own, I also converted the debiased between-group variance into a standard deviation on the original measurement scale. For birth weight, this yields a between-subpopulation standard deviation in grams; for maternal and paternal age, it yields a between-subpopulation standard deviation in years. Comparing these values across outcomes helps clarify whether birth weight varies across subpopulations to the same extent as the parental characteristics that may partly predict it.

To assess the stability of these heterogeneity estimates, I repeated the calculations after filtering out subpopulations with very small sample sizes. This checks whether the estimated between-group variation is driven mainly by sparse strata or whether it persists among larger and more stable subpopulations.

4 RESULTS

4.1 Maternal age is associated with birth weight, but the relationship is not purely linear

The first set of analyses shows that maternal age is an important predictor of birth weight, but its effect is not well summarized by a simple linear trend. In the individual-level analysis, maternal age and paternal age were both treated as plausible predictors, and the exploratory modeling framework allowed for nonlinear patterns in age effects. The resulting fitted relationships suggest that birth weight tends to be lower among very young mothers, increases over early adult ages, and then does not continue increasing at the same rate across the full age range. As shown in Figure 3, maternal and paternal ages are strongly associated in the observed birth records, with most births concentrated among parents in their 20s and 30s. This supports the decision to model parental age flexibly rather than forcing a constant linear effect.

This finding is substantively important because it suggests that birth weight risk is concentrated at specific parts of the maternal age distribution, rather than changing uniformly year by year. In particular, the analysis indicates that younger maternal age is associated with lower average birth weight, but the strength and shape of this association depend on how the population is stratified and whether other factors are controlled for. This motivates the later heterogeneity analysis, where the maternal-age effect is allowed to vary across subpopulations rather than being assumed constant nationwide.

4.2 Important birth weight variation remains across geographic and demographic subpopulations

After restricting attention to first-born singleton births, the meta-analysis aggregates births into subpopulations defined by county, year, sex, and maternal race. This yields 26,337 subpopulations with sufficient data to estimate subgroup means and standard deviations. As shown in Figure 1, subgroup sample sizes vary widely, so uncertainty adjustment is essential when comparing subgroup summaries.

Using the intraclass correlation coefficient (ICC), the analysis finds meaningful heterogeneity in mean birth weight across subpopulations. Across minimum stratum size thresholds, the birth weight ICC ranges from about 0.035 to 0.055, implying that roughly 4%–6% of total birth weight variation is attributable to between-subpopulation differences. Although smaller than the corresponding ICC for maternal age, this still indicates substantively important heterogeneity. Figure 4 shows that the birth weight ICC remains positive across thresholds, while maternal age consistently exhibits stronger between-subpopulation variation.

A complementary summary is the debiased standard deviation of subgroup means. For birth weight, this quantity is approximately 106–132 grams across thresholds, indicating that average birth weight still shifts meaningfully across counties and demographic strata even after accounting for estimation error.

4.3 Maternal age varies more strongly across subpopulations than birth weight

The same ICC analysis shows that maternal age exhibits stronger between-subpopulation heterogeneity than birth weight. The maternal-age ICC ranges from about 0.114 to 0.188, much larger than the birth weight ICC over the same thresholds. The notebook further translates this result into an interpretable scale by showing that the standard deviation of subpopulation mean maternal age is about 2.27 years at one representative threshold. Under a rough normal approximation, this implies that otherwise comparable subpopulations can differ by roughly 4.5 years in average maternal age between relatively low and high maternal-age strata. The contrast is also visible in Figure 4, where the maternal-age ICC remains well above the birth weight ICC across all subgroup size thresholds considered.

This matters for interpretation because it suggests that part of the observed variation in birth weight may reflect broader differences in population composition. In other words, subpopulations are not only different in birth outcomes; they also differ substantially in the age profile of mothers. This reinforces the need to distinguish compositional differences from within-subpopulation associations, and it supports the memo's broader focus on both individual predictors and subpopulation heterogeneity.

4.4 The association between maternal age and birth weight is positive on average among younger mothers, but it is heterogeneous

To study whether the maternal-age effect is consistent across contexts, the meta-analysis next focuses on mothers aged 30 or younger and estimates the Kendall rank correlation between maternal age and birth weight within each subpopulation. At the individual level, the overall Kendall's tau is approximately 0.073, indicating a positive association in this younger-mother subset. When the same quantity is estimated separately within subpopulations, the average tau is about 0.045. Thus, the positive age–birth-weight association persists on average, but it is weaker once examined locally rather than pooled over the entire population.

More importantly, the association is not constant across subpopulations. The raw standard deviation of the stratum-specific tau estimates is about 0.038, and after removing sampling variation the structural standard deviation remains about 0.019. This means the maternal-age association is genuinely heterogeneous across counties, years, sex groups, and maternal-race strata. Under a rough normal approximation, some subpopulations may have near-zero associations whereas others may show substantially stronger positive age effects. Therefore, the data do not support a single universal maternal-age slope; instead, the strength of association depends on local demographic and geographic context.

4.5 Evidence against a global null: many subpopulations show real age-related signal

The local false discovery rate analysis provides stronger evidence that the maternal-age association is not driven by a small number of isolated strata. After standardizing the stratum-specific tau estimates into Z-scores, the observed Z-score distribution is shifted to the right and more dispersed than a standard normal reference, as shown in Figure 6. Specifically, the mean Z-score is about 1.75 and the standard deviation is about 1.67, both inconsistent with the global null hypothesis that all subpopulation-specific associations are zero.

The quantile comparison and local FDR summaries lead to the same conclusion. The empirical Z-score curve lies above the standard normal reference over much of its range, and the observed Z-score distribution is shifted to the right and more dispersed than the null benchmark (Figure 6). Figure 2 further shows that local false discovery rates generally decrease with subgroup sample size, although evidence of signal is not confined to only the largest strata. Thus, evidence for a positive maternal-age signal appears across a broad range of subpopulations rather than being concentrated in a few high-count counties. Taken together, these results support the conclusion that birth weight variation in U.S. birth records reflects both individual-level predictors and meaningful heterogeneity in how those predictors operate across subpopulations.

5 FINDINGS AND DISCUSSION

The analyses show that birth weight variation in U.S. birth records is not well described by a single national pattern. At the individual level, maternal age is an important predictor, but its relationship with birth weight is not purely linear. As suggested by the parental age structure in Figure 3, age effects are better modeled flexibly than as a constant year-by-year change.

At the same time, this relationship is not uniform across counties, years, sex groups, and maternal-race strata. Figure 1 shows that subgroup sizes are highly uneven, while Figure 4 indicates that mean birth weight still exhibits meaningful between-subpopulation heterogeneity after accounting for estimation uncertainty. Maternal age varies even more strongly across subpopulations than birth weight, suggesting that part of the observed birth weight heterogeneity reflects broader compositional differences in the populations giving birth.

The subgroup-level Kendall correlation analysis reinforces this conclusion. Among mothers aged 30 or younger, maternal age is positively associated with birth weight on average, but the strength of this association differs across subpopulations. Figure 6 shows clear departure from the global null, and Figure 2 further suggests that this signal is widespread rather than driven only by a few extreme strata.

These results imply that pooled national summaries may conceal meaningful local variation. They also show the value of combining individual-level regression with subpopulation-level meta-analytic summaries: the first identifies major predictors and nonlinear structure, while the second reveals whether those patterns are stable across contexts.

Several limitations remain. This study is observational, so the estimated relationships should not be interpreted causally. Maternal age may also proxy for unmeasured social, behavioral, or medical factors. In addition, the subgroup analysis is based on aggregated summaries, so it is most informative about heterogeneity in average patterns rather than the full individual-level mechanism.

Therefore, the main conclusion is that birth weight variation reflects both individual predictors and substantial subpopulation heterogeneity. Maternal age matters, especially at younger ages, but its association with birth weight is neither purely linear nor constant across contexts.

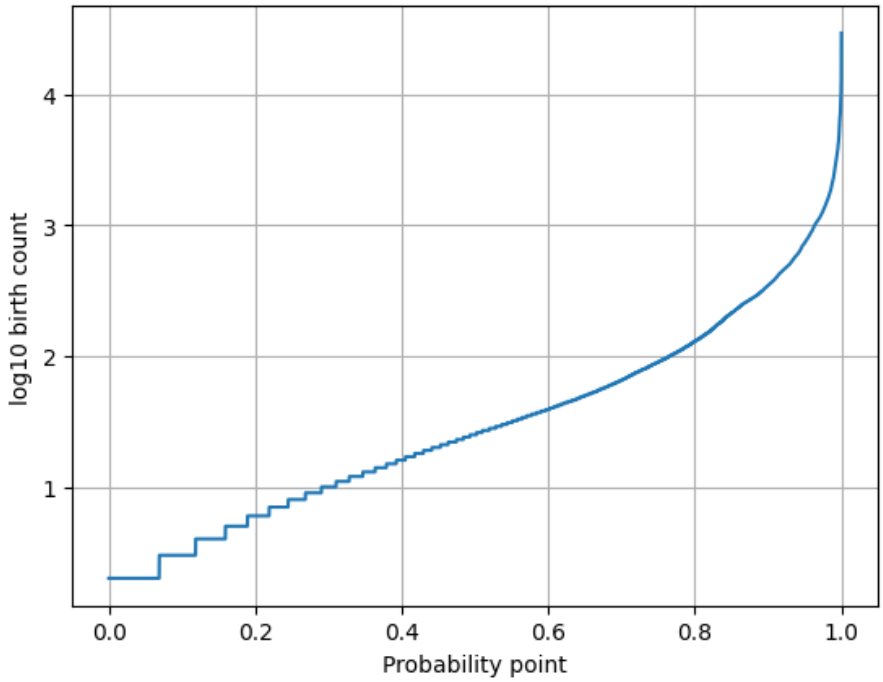

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Figure 1. Empirical quantile plot of subgroup sample sizes for the county–year–sex–maternal-race strata used in the subpopulation analysis. The broad spread of subgroup sizes indicates substantial imbalance across strata, with many small groups and a smaller number of very large groups. This unevenness motivates the use of uncertainty-adjusted summaries when comparing subgroup means and assessing heterogeneity.

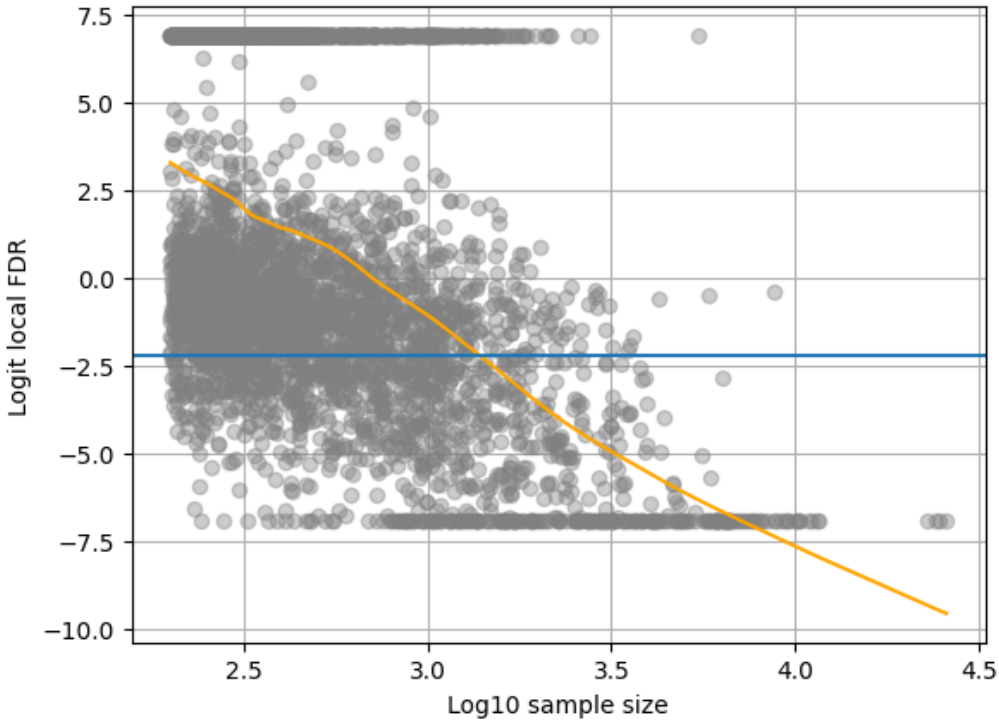

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Figure 2. Relationship between subgroup sample size and local false discovery rate for the stratum-specific association between maternal age and birth weight. Each point corresponds to a subgroup, with the vertical axis shown on the logit scale. The smoothed trend declines with sample size, indicating that larger strata tend to provide stronger evidence against the null, although low local false discovery rates are also observed in some smaller strata. The horizontal reference line corresponds to a local false discovery rate threshold of 0.1 on the logit scale.

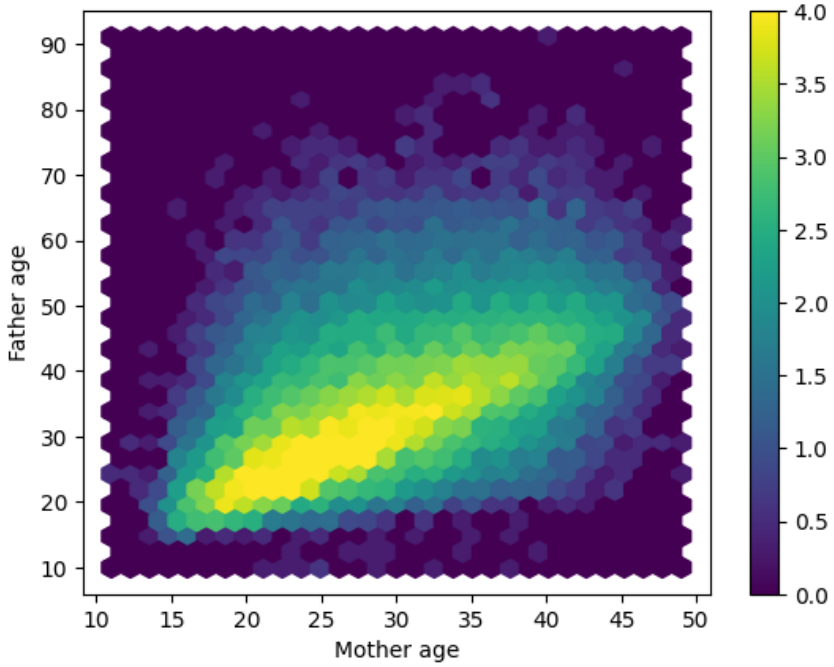

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Figure 3. Hexbin plot of maternal and paternal age in the individual-level birth records. Maternal and paternal ages are positively associated, with the highest concentration of births occurring among parents in their 20s and 30s. The broad but structured age distribution supports modeling parental age as an important predictor of birth weight and motivates the use of flexible, nonlinear age effects in the regression analysis.

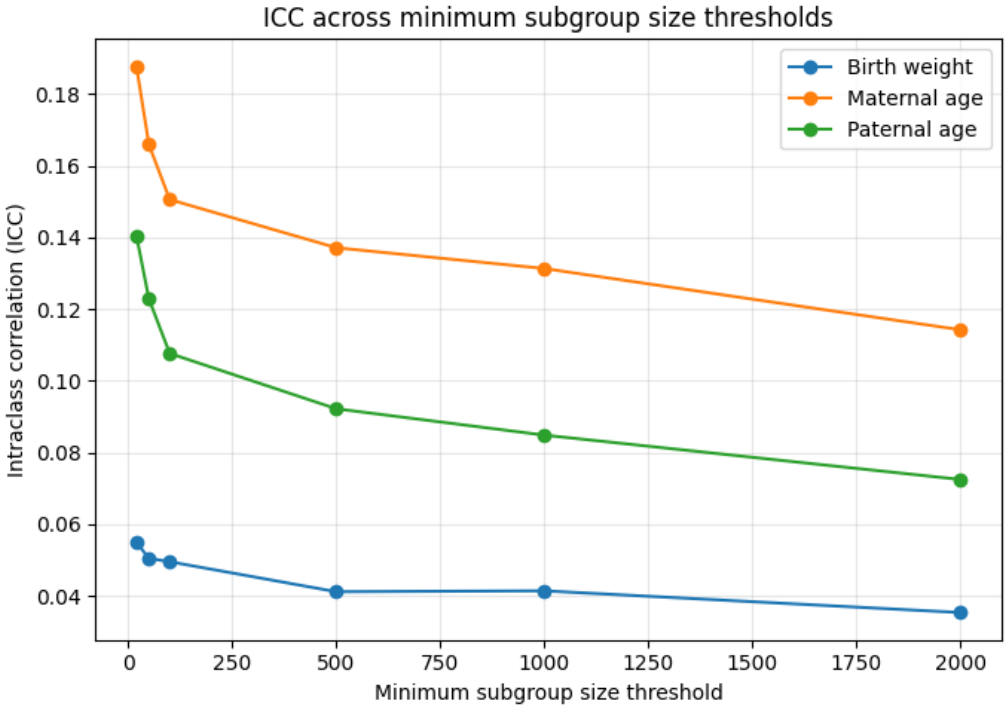

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Figure 4. Estimated intraclass correlations (ICCs) for birth weight, maternal age, and paternal age across different minimum subgroup size thresholds. All three quantities show persistent between-subpopulation heterogeneity, but maternal age exhibits consistently stronger heterogeneity than birth weight. The birth weight ICC remains positive across thresholds, indicating that subgroup-level differences persist even after excluding sparsely populated strata.

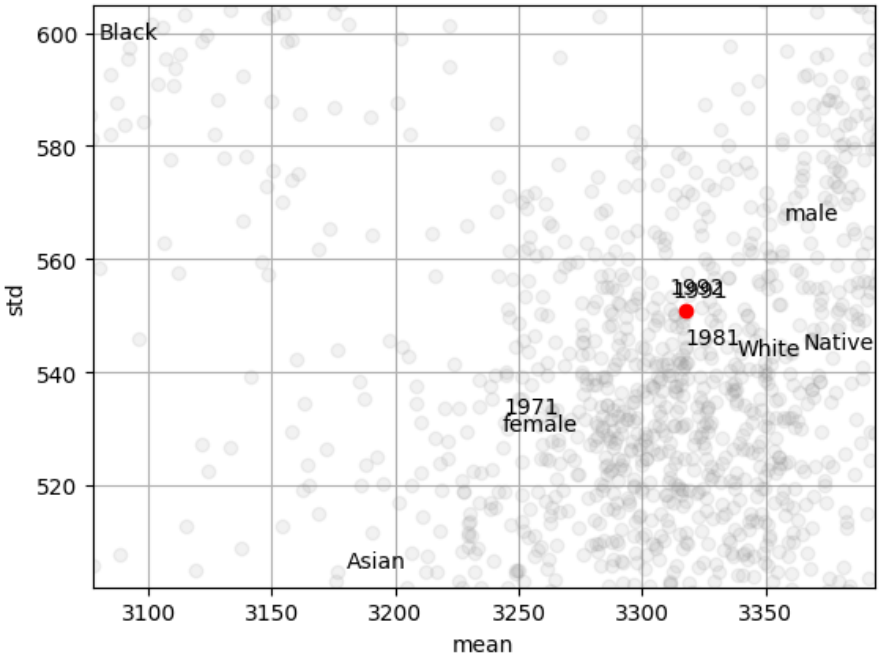

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Figure 5. Zoomed-in scatterplot of subgroup mean birth weight versus subgroup standard deviation, with selected subpopulations annotated by year, sex, or maternal race. Even within a relatively narrow range of subgroup means, substantial variation in dispersion remains visible, indicating that subgroup heterogeneity is not limited to differences in average birth weight alone. The annotated points highlight how demographic and temporal strata occupy different locations within this mean–standard deviation relationship.

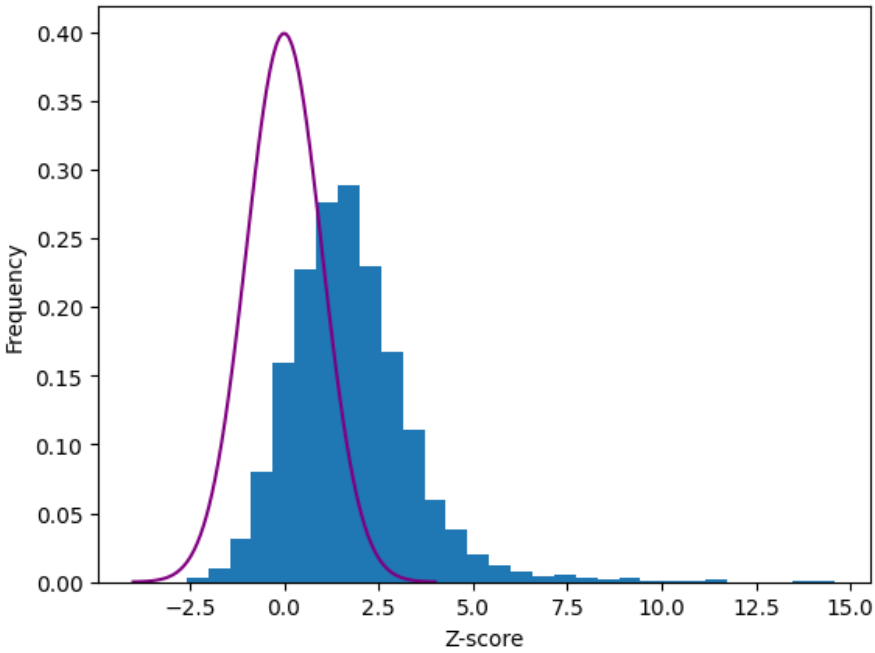

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Figure 6. Distribution of stratum-specific Z-scores for the subgroup-level association between maternal age and birth weight, with the standard normal density overlaid as a reference. Relative to the null reference, the observed distribution is shifted to the right and is more dispersed, indicating that many subpopulations show evidence of a positive maternal-age signal. This pattern is inconsistent with a global null hypothesis in which all subgroup-specific associations are zero.